

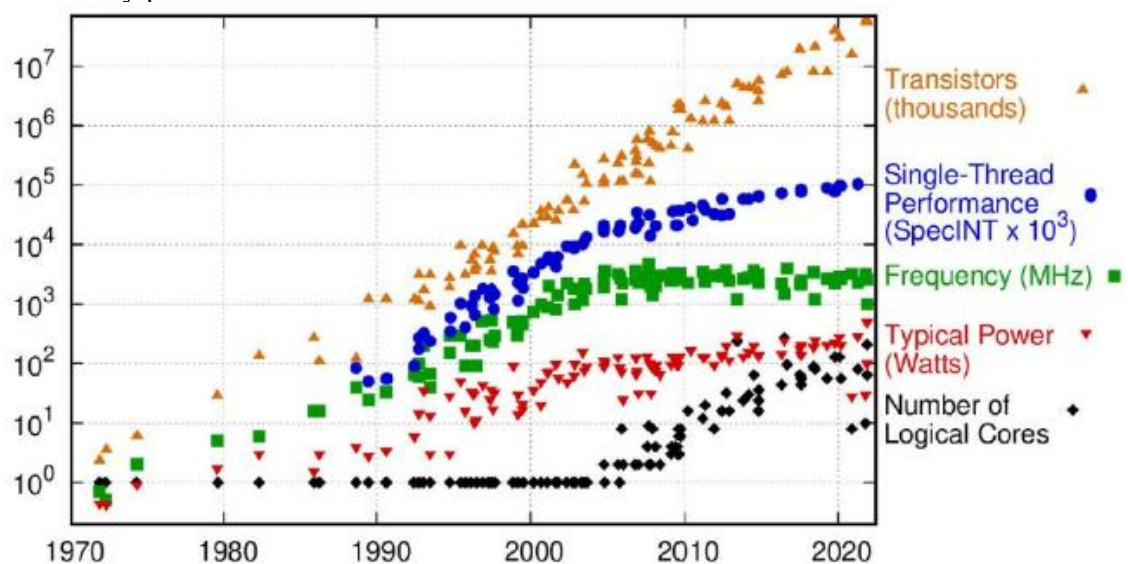
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Parallel Computing - IT3243

Tutorial No 02: Computer Organization & Trend in Parallel Architecture

1. Define the following:
 - a. *clock frequency*
 - b. *power consumption*
 - c. *performance*
2. Define the following terms relates with the **relative performance of a computer**:
 - a. *Latency*
 - b. *Throughput*
3. State that does the following changes to a computer system increase throughput, decrease latency, or both?
 - a. *Replacing the processor in a computer with a faster processor.*
 - b. *Adding additional processors to a system that uses processors for separate tasks.*
4. State that a machine X is n times faster than a machine Y. What does this mean?
5. If machine X takes 20 seconds to perform a task and machine Y takes 2 minutes to perform the same task, then machine X is how many times faster than machine Y?
6. Computer C's performance is 4 times faster than the performance of computer B, which runs a given application in 28 seconds. How long will computer C take to run the application?
7. Define the following that relates with the **measures of the execution time** on a computer:
 - a. *Clock cycle*
 - b. *Clock period*
 - c. *Clock rate*
8. Define the **CPU Execution time** in terms of *clock cycles* and *clock rate*.
9. A favorite program runs in 10 seconds on computer A, which has a 2 GHz clock. We are trying to help a computer designer build computer B, which will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but it will affect the rest of the CPU design, causing computer B to require 1.2 times as many clock cycles as computer A for this program. What clock rate should we tell the designer to target?
10. Define the **CPU Execution time** in terms of *instructions performance*.

11. Suppose we have two implementations of the same instruction set architecture. Computer A has a clock cycle time of 250 ps and a CPI of 2.0 for some program, and computer B has a clock cycle time of 500 ps and a CPI of 1.2 for the same program. Which computer is faster for this program and by how much?
12. Define the classic **CPU time** in terms of instruction count, CPI, and clock cycle time.
13. State *three* aspect of computer design.
14. Discuss the architectural ideas that have been applied in the design of computers for over half a century now.
15. State clearly the low posits by the following computer scientist in parallelism :
 - a. Amdahl's low
 - b. Moore's low
 - c. Dennard's low- Dennard Scaling
16. Consider the following figure shows the trends in clock frequency, power consumption, the number of computational cores, and hardware performance over time for commodity processors.



- a) What would be the significant change on year 2005?
- b) Explain the concept of the 'power wall' and its relation to the trend in clock frequencies.
- c) Explain how this trend in microprocessor clock rates impacts the approach to designing parallel algorithms.
- d) Describe the trend concerning the number of transistors and the corresponding evolution in transistor size, and forecast the future implications of this trend.